Notes: Calomiris & Kahn (AER, 1991)

The Role of Demandable Debt in Structuring Optimal Banking Arrangements

Contents

1 Big picture

- **Question.** Why did banks historically finance themselves mainly with demandable debt - notes and deposits payable on demand - despite the obvious costs of maturity mismatch, reserves, suspension, receivership, and liquidation?
- **Core answer.** Demandable debt is an incentive device. It gives depositors an option to withdraw and thereby force liquidation, which disciplines a banker who has superior information and investment expertise but may act fraudulently against uninformed depositors.
- **Key institutional puzzle.** Liquidity-based theories explain why depositors may want flexible consumption, but they do not explain why individual banks were forced into receivership rather than merely suspending convertibility, nor why first-come-first-served service mattered so much.
- **Historical motivation.** The authors emphasize that bank failures often involved fraud or conflicts of interest, and that informed depositors and other banks often triggered runs based on information about bank asset values.
- **Sequential service.** First-come-first-served repayment is not an accidental feature. It rewards informed monitors for acting early. Monitors who receive bad signals can withdraw before uninformed depositors and thus have an incentive to produce information.
- **Liquidation as commitment.** The banker's willingness to be liquidated under certain states lowers his cost of raising funds because it commits him not to abscond with assets after bad realizations.
- **Single-depositor model.** If the depositor can buy a signal about the bank's future return, the optimal contract can be a contingent liquidation contract: liquidate after a bad signal, do not liquidate after a good signal.
- **Multiple-depositor model.** With many potential monitors receiving independent signals, the bank sets a threshold number of bad signals. If withdrawals exceed the threshold, the bank is liquidated; otherwise it continues.
- **Role of reserves.** Reserves are not only for transactions or portfolio diversification. They help implement the incentive scheme by allowing early withdrawing monitors to be paid without forcing liquidation after only a few bad signals.
- **Liquidity/transactability.** The fact that the bank remains open conveys information: enough monitors have not run. This makes demandable bank liabilities easier for nonmonitors to value and hence more transactable.
- **Main takeaway.** Runs and liquidation are not merely bad equilibrium accidents. In this model, the threat of a run is part of the governance technology that makes bank intermediation possible.

2 Data and historical motivation (banking institutions and observables)

2.1 Why demandable debt needs an explanation

- Demandable debt created a maturity mismatch: banks held illiquid assets but issued liabilities payable immediately.
- This mismatch exposed banks to costly individual-bank liquidation, system-wide suspension, and reserve holding.
- At first glance, maturity-matched debt or equity seems cheaper because it avoids forced liquidation and idle reserves.
- The paper argues that this view misses the incentive value of withdrawal rights.

Interpretation. The contract looks costly only if we ignore the banker's moral hazard. Once the banker can abscond or divert assets, demandability has a governance function.

2.2 What standard liquidity theories miss

- Diamond-Dybvig-style theories emphasize depositors' uncertain consumption timing.
- These theories explain why demand deposits can provide liquidity, but they treat runs as unfortunate side effects.
- Calomiris and Kahn argue that historical individual-bank liquidation was not just a liquidity accident.
- Suspension of convertibility was available for systems during panics, but individual troubled banks were placed into receivership.

Interpretation. The key historical fact is not just that depositors sometimes withdrew. It is that withdrawal orders transferred control away from the banker.

2.3 Historical facts the model targets

- Bank failures often involved fraud, insider loans, defalcation, or related-party lending.
- Informed depositors, including other bankers, often learned of trouble before ordinary depositors.
- Runs by informed depositors depleted reserves and forced receivership.
- First-come-first-served repayment gave early withdrawers priority.
- Payments made shortly before bankruptcy were normally suspect outside banking, but in banking early withdrawers often did receive preferential treatment.

Interpretation. These facts motivate a theory in which informed early withdrawals are socially useful because they force liquidation before the banker can steal or dissipate assets.

2.4 Actors in the model

- **Banker:** has a profitable investment opportunity and a comparative advantage in allocating funds, but may abscond after returns are realized.
- **Depositors:** provide funds and require at least the outside expected return S.
- **Monitors:** depositors who can pay a cost I to receive a signal about the bank's return.
- **Nonmonitors:** depositors for whom producing information is too costly; they free ride on monitors but accept lower payoffs in bad states.
- **Receiver/court:** can take control of assets if liquidation is mandated.

Interpretation. The banking relationship is three-sided: banker, monitoring depositors, and passive depositors. Demandable debt coordinates these three roles.

2.5 The two central institutional features

- **Withdrawal option:** gives depositors the right to force liquidation when they see bad information.
- **Sequential service:** gives monitors priority repayment, compensating them for information production.

Interpretation. Without sequential service, monitoring would be a public good and depositors would free ride. Without liquidation rights, monitors could not discipline the banker.

3 Models and theoretical specifications (all equations plus interpretation)

3.1 Single-depositor physical structure

A banker needs one dollar to fund an investment. The investment returns

$$T_i \in \{T_1, T_2\}, \qquad T_2 > T_1,$$

with high outcome probability γ. The realization is not contractible because it is observed ex post only by the banker. The depositor's outside expected return is S.

Interpretation. The model begins with a classic costly-state-verification problem: the banker knows the realized return, but the depositor and court cannot contract directly on it.

3.2 Absconding

Immediately before repayment, the banker may abscond. If he absconds, the value of bank assets is reduced by a fraction governed by A. A promise to pay P is credible only if the banker prefers payment to absconding. In the paper's notation, the banker compares the promised payment with the absconding tax:

$$\text{abscond if } P > AT_i.$$

Interpretation. Low realizations make absconding more tempting because the amount the banker loses by stealing is smaller.

3.3 Why default premia alone may not work

If the promised repayment must include a default premium, the premium itself can make absconding more attractive. The paper's example highlights this case:

$$S > AT_1,$$

so any promised payment large enough to attract the depositor is high enough to trigger absconding in the low state. Even if the project is socially desirable,

$$\gamma T_2 + (1-\gamma)(1-A)T_1 > S,$$

it may still be impossible to fund if

$$S > \gamma A T_2.$$

Interpretation. A simple promised payment can collapse under incentive constraints: paying enough to attract funds can simultaneously make fraud attractive.

3.4 Liquidation technology

A receiver can take the assets away from the banker in period 2. Liquidation is costly, but it preempts the banker's ability to abscond. The paper assumes a maximum liquidation payment M satisfying

$$AT_2 > M > AT_1,$$

and imposes the maintained liquidation-cost condition

$$L > A.$$

Interpretation. Liquidation is a costly commitment technology. It is not used because receivers are efficient operators; it is used because receivers can place assets beyond the banker's control.

3.5 Private signal and demandable-debt logic

The depositor may pay cost I in period 1 to receive a signal

$$\sigma \in \{g, b\},$$

where

$$p_g > \gamma > p_b.$$

The depositor can then announce a signal and the contract maps the announcement into repayment and liquidation.

Interpretation. Monitoring is costly but useful. A bad signal makes liquidation more attractive because it prevents expected absconding in low-return states.

3.6 Contract types

A contract maps announcements into outcomes (P, Λ) where P is the promised repayment and $\Lambda \in \{0, 1\}$ indicates liquidation. With one depositor and a binary signal, the relevant contracts are:

- simple nonliquidating contract;
- simple liquidating contract;
- compound nonliquidating contract with different payments after good/bad signals;
- demandable debt: liquidate after a bad signal and do not liquidate after a good signal.

Interpretation. Demandable debt is the contract in which the depositor's bad signal becomes a liquidation demand.

3.7 Single-depositor characterization

The paper's Theorem 2 says that if the signal is cheap and accurate, the optimal contract changes with the required return S:

- for low S, simple nonliquidating debt is optimal;
- for intermediate S, a "nuisance" compound nonliquidating contract may be optimal;
- for higher but still feasible S, demandable debt is optimal;
- for too high S, no contract is feasible.

Interpretation. Demandable debt is most useful when depositors require enough compensation that simple debt would create a serious absconding problem.

3.8 Multiple-depositor structure

There are Z potential depositors and K potential monitors. The project costs Y and returns YT_i. Deposits in excess of Y are held as reserves and earn the competitive return S.

Let N be the number of monitors who receive and announce the bad signal. Conditional on the state, signals are i.i.d., and N is a sufficient statistic for the likelihood of the low state.

Interpretation. More monitors improve information aggregation. The bank then uses a threshold rule: a few bad signals are absorbed by reserves; many bad signals trigger liquidation.

3.9 Aggregate contract constraints

The aggregate contract must satisfy three conditions:

1. Depositors receive their aggregate reservation payoff plus monitoring costs:

$$SZ + KI.$$

2. If liquidation occurs, payment cannot exceed the feasible liquidation value:

$$P \leq MY + (Z - Y)S.$$

3. If liquidation does not occur, the banker absconds whenever

$$AT_iY < P - (Z - Y)S.$$

Interpretation. The reserves term $(Z - Y)S$ matters because reserves can be returned without relying on the banker's private project.

3.10 Threshold liquidation rule

Theorem 3 says that for sufficiently high but feasible required returns, the optimal contract has a cutoff $\underline{N}$:

$$N > \underline{N} \Rightarrow \Lambda(N) = 1, \qquad P(N) = MY + (Z - Y)S,$$
$$N < \underline{N} \Rightarrow \Lambda(N) = 0, \qquad P(N) = AT_2Y + (Z - Y)S.$$

Interpretation. If many monitors report bad signals, the bank is liquidated and depositors receive a safe liquidation payoff. If few do, the bank continues and promises a high payoff, but absconding can occur if the project is actually bad.

3.11 Standard demandable-debt implementation

A standard demandable-debt contract pays any depositor who announces b a certain amount R. Announcing b is interpreted as withdrawal. If more than $\underline{N}$ depositors withdraw, the bank is liquidated; otherwise it continues. The individual incentive constraints reduce to

$$EU(\hat{g}, b) = R - \frac{I}{1 - \lambda},$$
$$S > R,$$

where λ is the prior probability of a good signal.

Interpretation. The payment R must be high enough to compensate monitors for information costs when they truthfully withdraw after bad signals, but not so high that nonmonitors prefer to always withdraw.

3.12 Reserves and sequential service

Reserves allow the bank to pay a small number of early withdrawers with certainty. This lets the bank reward monitors without automatically forcing receivership after every isolated bad signal.

Interpretation. The model adds an incentive-compatibility motive for reserves. Reserves are not just liquidity buffers; they make sequential service credible.

3.13 Transactability

When the bank remains open, nonmonitors learn that monitors have not seen enough bad signals to force liquidation. This information places a lower bound on the value of bank liabilities.

Interpretation. Demandability can make bank liabilities more transactable because the monitoring system reveals information through the absence of a run.

4 Identification and theoretical design (what makes the argument credible)

4.1 Theoretical identification rather than econometric identification

This is a theoretical paper, not an empirical event study. Its credibility comes from explaining institutional features that standard liquidity theories struggle to explain:

- individual-bank liquidation rather than discretionary suspension;
- fraud and insider abuse in historical bank failures;
- information-based runs by informed depositors;
- first-come-first-served repayment;
- reserves despite their opportunity cost;
- the high transactability of demandable bank liabilities.

Interpretation. The model is persuasive because it turns apparently costly or puzzling banking features into components of an incentive-compatible contract.

4.2 Why liquidation can be efficient ex ante

- Ex post, liquidation is costly.
- Ex ante, liquidation rights make depositors willing to fund the bank.
- Without the liquidation option, the banker may have no credible way to promise repayment.
- Therefore, costly liquidation may expand the set of feasible investments.

Interpretation. The relevant comparison is not liquidation versus no liquidation after funding has occurred. It is demandable debt with feasible intermediation versus no funding because the banker's promises are not credible.

4.3 Why first-come-first-served matters

- Monitoring is costly and its benefits partly accrue to everyone.
- Sequential service rewards monitors because they can withdraw early after bad signals.
- This reward makes monitoring privately worthwhile.
- Passive depositors accept being later in line because monitors discipline the banker.

Interpretation. Sequential service solves a free-riding problem in information production.

4.4 Why multiple monitors and reserves matter

- Individual signals can be noisy.
- Aggregating independent signals improves inference about bank quality.
- A threshold rule prevents unnecessary liquidation after only a few bad signals.
- Reserves buffer early withdrawals and make the threshold rule workable.

Interpretation. The bank can separate isolated monitor withdrawals from a true information-based run.

4.5 What the paper cannot fully prove

- The model is stylized and uses a simplified “absconding” technology.
- It does not model active trading in bank equity, which could provide alternative discipline in modern markets.
- It focuses on individual banks in normal times, not system-wide panics.
- It intentionally abstracts from depositors’ demand for consumption liquidity.
- It does not fully model the transactions demand for bank liabilities; it only sketches the information-sharing channel.

Interpretation. The paper’s main contribution is not a complete banking model. It is a theory showing that demandability can be valuable because it disciplines bankers.

5 Tables and figures

The original paper has two formal tables and no standalone empirical figures. To preserve the paper's visual material while keeping the notes compact, this section directly crops and groups the original tables plus key theorem/equation panels from the paper.

5.1 Theorem 1 + Theorem 2: when demandable debt is optimal

Read:

- Theorem 1 classifies the possible optimal contracts in the single-depositor model.
- Demandable debt is the compound contract with liquidation after a bad signal and continuation after a good signal.
- Theorem 2 says demandable debt arises when the signal is cheap and accurate and the depositor's required return is sufficiently high.
- If S is low, simple nonliquidating debt is enough. If S is too high, no contract is feasible.

Economic message. Demandable debt is not always optimal. It becomes optimal when the banker's moral-hazard problem is severe enough that simple promises are not credible, but monitoring and contingent liquidation are still useful.

THEOREM 1:[17] *The optimal contract in the problem takes one of the following four forms*:
a) *a simple nonliquidating contract*
b) *a simple liquidating contract*; *in this case*,

$$AT_1 < P \leq M$$

c) *a compound contract composed of two simple nonliquidating contracts* ($\Lambda_b = \Lambda_g = 0$); *in this case*,

$$P_b \leq AT_1 \text{ and } AT_1 < P_g \leq AT_2$$

d) *a compound contract composed of one simple liquidating contract and one simple nonliquidating contract* ($\Lambda_b = 1, \Lambda_g = 0$); *in this case*,

$$AT_1 < P_b < P_g \leq AT_2.$$

Theorem 1: contract forms

THEOREM 2: *If the signal is sufficiently cheap and accurate*, *then there exist values* S^* *and* $\hat{S}$, *such that the optimal contract depends on the required returns* S *in the following way*: *for* $S \leq AT_1$ *the simple*, *nonliquidating contract is optimal*; *for* $S \in (AT_1, S^*]$, *the nuisance contract is optimal*; *for* $S \in (S^*, \hat{S}]$, *demandable debt is optimal*; *and for* $S > \hat{S}$, *no contract is feasible*.

Theorem 2: required-return regions

5.2 Table 1 + Table 2: payoff accounting under multiple depositors

Read:

- Table 1 lists payoffs under three nodes: liquidation, no liquidation with absconding, and no liquidation without absconding.
- Under liquidation, depositors receive P and the banker receives the residual after liquidation and reserves.
- If the banker absconds, depositors keep only reserves; the banker takes the project payoff net of the absconding loss.

- Table 2 shows the payoff to a depositor who announces g when other depositors' bad-signal announcements are below or above the liquidation threshold.
- The table makes clear that early withdrawers receive the fixed withdrawal payoff R, while nonwithdrawers split the remaining aggregate payoff.

Economic message. The payoff tables are the accounting backbone of the model. They show how reserves, liquidation, and sequential service jointly implement the incentive scheme.

TABLE 1—PAYOFFS ON EACH OF THE THREE NODES OF THE GAME TREE

Contract	Banker receives	Depositors receive
Liquidation	$(1-L)T_iY+(Z-Y)S-P$	P
No liquidation		
Banker absconds	$(1-A)T_iY$	$(Z-Y)S$
Banker does not abscond	$T_iY+(Z-Y)S-P$	P

Table 1: payoffs at the game-tree nodes

TABLE 2—PAYOFF TO DEPOSITOR WHO ANNOUNCES g

	Payoff to depositor announcing g	
Project realization	Number of depositors announcing b < $\underline{N}$	Number of depositors announcing b > $\underline{N}$
T_1	$\frac{AT_2+(Z-Y)S-RN}{Z-N}$	$\frac{MY+(Z-Y)S-RN}{Z-N}$
T_2	$\frac{(Z-Y)S-RN}{Z-N}$	$\frac{MY+(Z-Y)S-RN}{Z-N}$

Table 2: payoff to a depositor announcing g

5.3 Theorem 3 + Theorem 4: threshold runs and standard demandable debt

Read:

- Theorem 3 characterizes the aggregate threshold rule in the multi-depositor model.
- If the number of bad signals is high, liquidation occurs and depositors receive the maximum feasible liquidation payoff.
- If the number of bad signals is low, liquidation does not occur; the promised payment is high, but absconding can occur in the bad productivity state.
- Theorem 4 says the optimal outcome can be implemented with a simple demandable-debt contract under the distributional assumptions.
- The cropped Theorem 4 panel also shows the individual incentive conditions that keep monitors truthful.

Economic message. A bank run is not just panic. In the model it is a decentralized trigger for liquidation when enough monitors receive bad information.

rate of return exceeds S, no contract is feasible. $\hat{S}$ can be calculated explicitly.

Our first result is that, for required returns which are sufficiently high (but less than $\hat{S}$), the optimal contract calls for liquidation when the number of bad signals is high, and not when the number of bad signals is low. When the number of bad signals is low, there is a positive (but small) probability that the banker will abscond.

THEOREM 3: *For an interval of values of S, $(\underline{S}, \hat{S}]$, the optimal contract has the following form: there exists $\underline{N}$ such that*:

If $N > \underline{N}$, $\Lambda(N) = 1$ *and*

$$P(N) = MY + (Z - Y)S;$$

If $N < \underline{N}$, $\Lambda(N) = 0$ *and*

$$P(N) = AT_2Y + (Z - Y)S.$$

Theorem 3: liquidation threshold

$Z - N$, yielding the quantity in the right-most column of the table. The remaining numbers are calculated in a similar fashion.

Given the probabilities of the realizations of T_i and the probability of each signal contingent on T_i, it is a straightforward matter to calculate $\mathrm{EU}(\hat{g}, \mathrm{b})$ and $\mathrm{EU}(\hat{g}, \mathrm{g})$. For this scheme, the incentive and participation constraints reduce to the following:[24]

$$\mathrm{EU}(\hat{g}, \mathrm{b}) = R - I/(1 - \lambda)$$
$$S > R.$$

When an aggregate contract of the sort described in the previous section is optimal, it can always be implemented with a demandable-debt scheme, as stated in the following theorem.

THEOREM 4: *Under the distributional assumptions and the conditions of the previous theorem, the optimal outcome can be achieved with a simple demandable-debt contract.*

Theorem 4: implementation by demandable debt

5.4 Transactability and summary mechanism

Read:

- The paper argues that the fact "the bank is open" is informative to nonmonitors.
- In the one-monitor case, no run is fully revealing that the monitor did not see the bad signal.
- In the multi-monitor case, no run is partially revealing: fewer than the liquidation threshold of bad signals has occurred.
- This information can make demandable bank claims more liquid and more useful in transactions.

Economic message. The transactability of demandable debt may be a by-product of the incentive system. Monitoring produces information; demandability shares that information with nonmonitors.

6 Short summary

- The paper asks why demandable debt became the central form of bank finance despite its costs.
- Its answer is that demandable debt disciplines bankers by giving depositors a liquidation option.
- The historical motivation is that bank failures often involved fraud and information-based runs, not just exogenous liquidity needs.
- Individual-bank liquidation is central to the contract because it removes assets from the banker's control.
- Sequential service gives informed monitors an incentive to run early when they receive bad signals.
- With one depositor, demandable debt can be optimal when the depositor's signal is cheap and accurate and the required return is high.
- With many depositors, independent signals are aggregated through the number of withdrawals.
- The optimal contract liquidates the bank only when bad-signal announcements exceed a threshold.
- Reserves allow the bank to pay a small number of early withdrawers without immediate liquidation.
- Passive depositors benefit from monitors because monitoring keeps the banker in line.
- Demandable debt also improves transactability because the absence of a run communicates information about the bank's condition.
- The paper does not model system-wide panics, deposit insurance, equity trading, or liquidity demand in full.
- The key lesson is that bank runs and liquidation rights can be part of an efficient governance arrangement, not merely inefficient breakdowns.

7 Conclusion

7.1 Core conclusion

Calomiris and Kahn argue that demandable debt can be an optimal banking arrangement because it makes intermediation incentive compatible. It allows depositors to discipline a banker who may otherwise abscond with assets or act fraudulently. The threat of liquidation, triggered by withdrawals, makes funding possible in settings where simple promises to repay would not be credible.

7.2 Economic intuition

The intuition is that withdrawal is a vote of no confidence. Monitoring depositors observe private signals. If enough of them see bad signals, they withdraw, and the bank is liquidated before the banker can steal or dissipate assets. Sequential service rewards these monitors for paying information costs. Reserves then make the system work by paying a small number of early withdrawers without forcing liquidation too often.

7.3 Takeaway

The paper reframes demandable debt. Its fragility is not only a cost; it is also the source of discipline. A bank that can be run can credibly promise depositors that bad information will remove control from the banker. This idea helps explain bank notes, deposits, receivership, sequential service, reserves, and the liquidity of bank liabilities in one unified contracting framework.